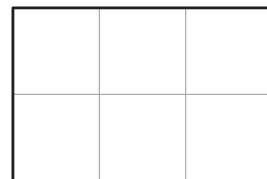
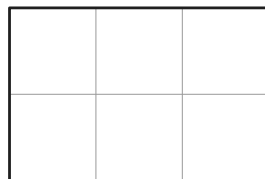
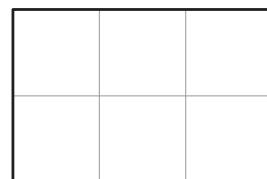
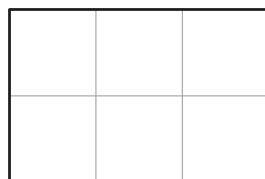
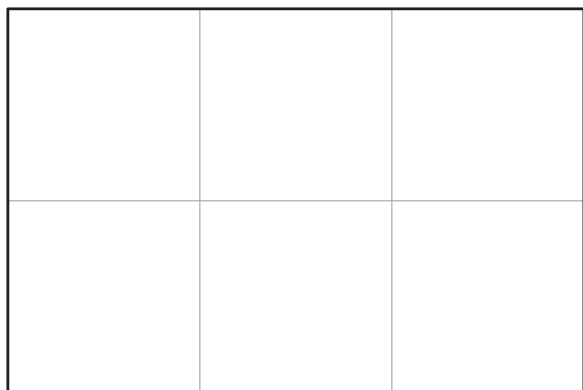
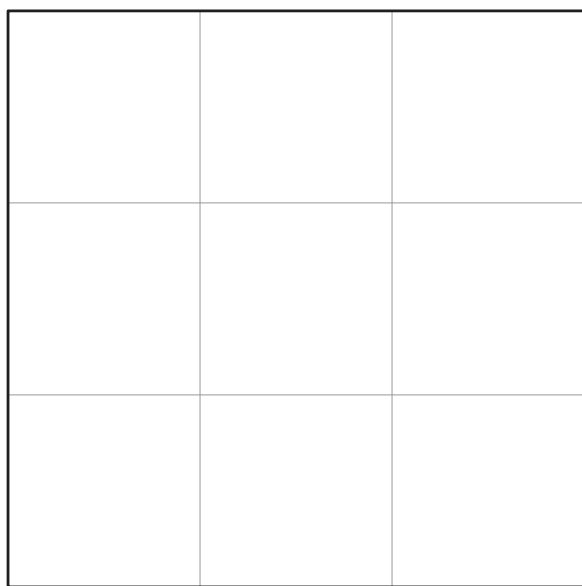


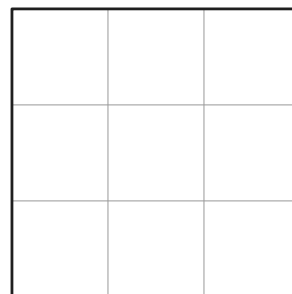
Problem 1: A domino covers two squares that share a side. Cover the 2-by-3 board with dominoes. Draw your way on a small board by drawing a thick line around each domino. Find every different way. Use a new small board for each way.



Problem 2: Put a black square on one square of the 3-by-3 board. Try to cover all the other squares with dominoes. If you can, draw a check on that square of the small map. If you cannot, draw an X there. Do this for all nine squares.

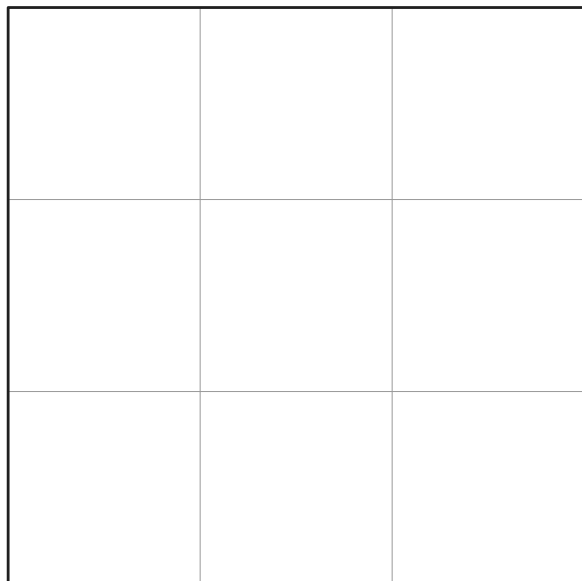


board



map

Problem 3: Shade the squares of this board like a checkerboard, starting with the top left square shaded. Write how many squares are shaded and how many are not. Compare the shaded squares with the checks on your map from Problem 2. Then put a domino on the board in three different places. For each place, write how many shaded squares and how many unshaded squares it covers. Use your counts to say why a black square on an unshaded square makes covering impossible. Say it to the adult at your table.



shaded squares _____

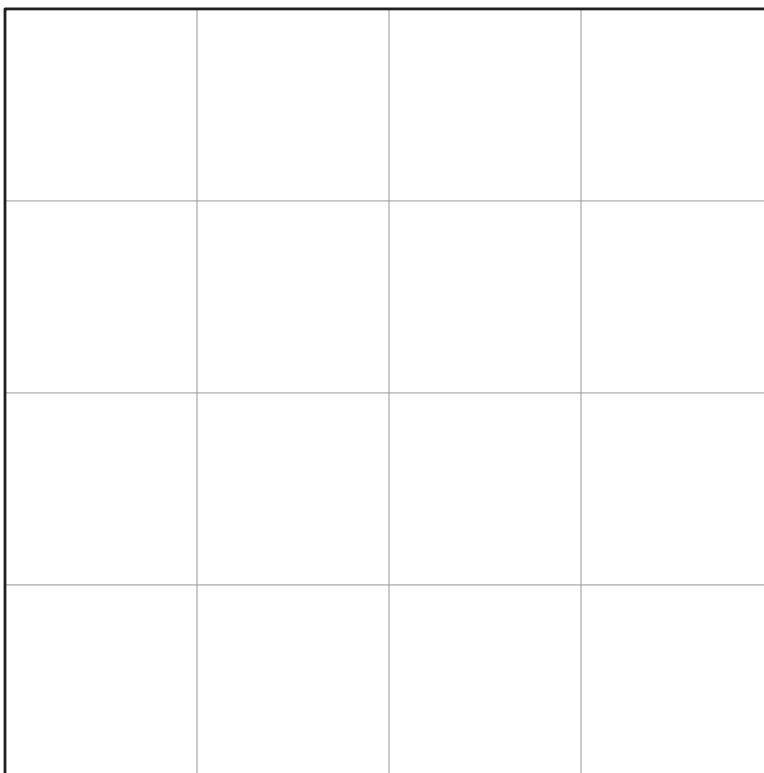
unshaded squares _____

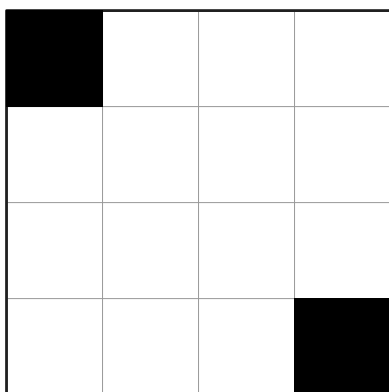
domino 1 covers shaded _____ unshaded _____

domino 2 covers shaded _____ unshaded _____

domino 3 covers shaded _____ unshaded _____

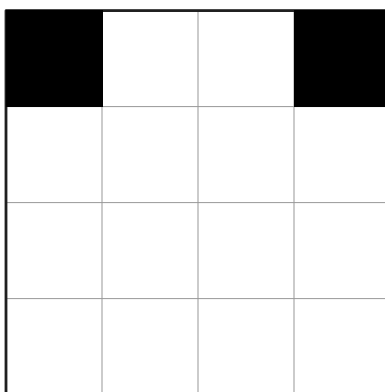
Problem 4: Each small board on the next page shows two black squares. For each small board, put two black squares in the same places on the big board below and try to cover all the other squares with dominoes. If you can, draw the dominoes on the small board. If you cannot, write NO under it.





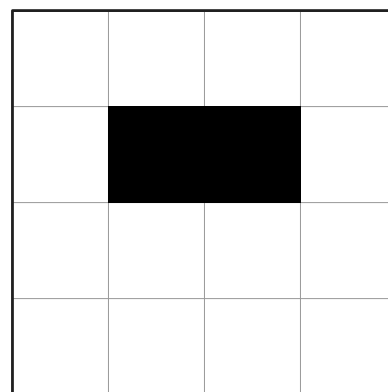
a

shaded _____ unshaded _____



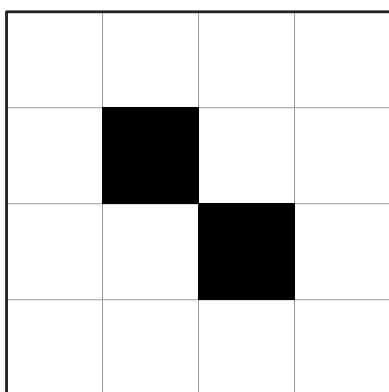
b

shaded _____ unshaded _____



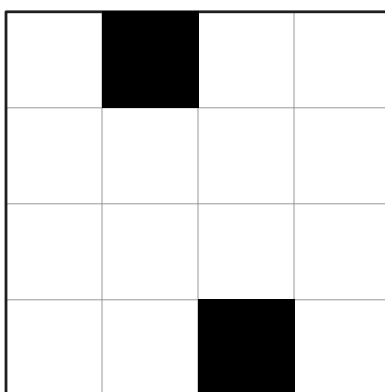
c

shaded _____ unshaded _____



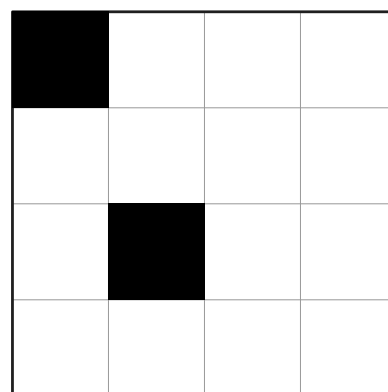
d

shaded _____ unshaded _____



e

shaded _____ unshaded _____

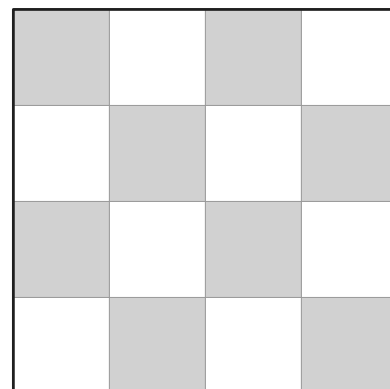
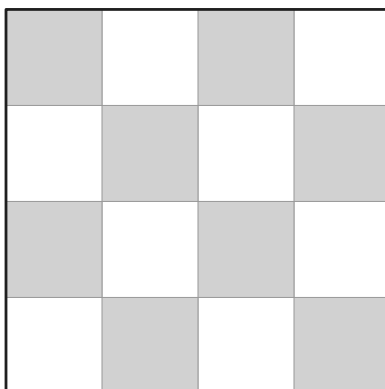
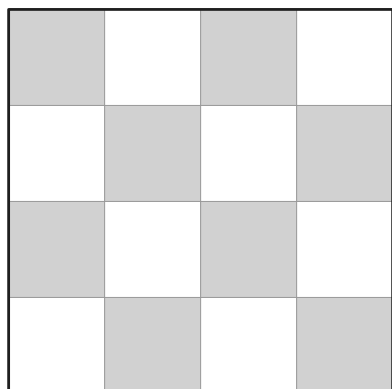


f

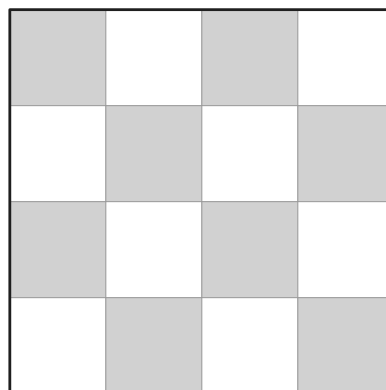
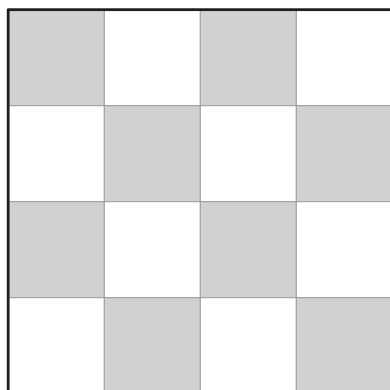
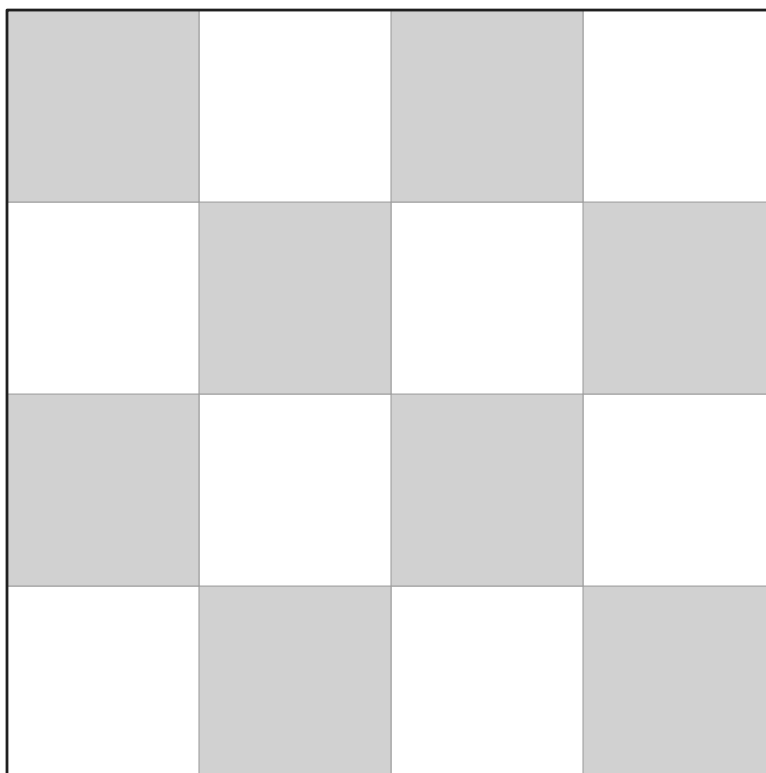
shaded _____ unshaded _____

Problem 5: Shade each of the six small boards like a checkerboard, starting with the top left square shaded. Under each board write how many shaded squares and how many unshaded squares are not black. Compare these numbers with your NO boards.

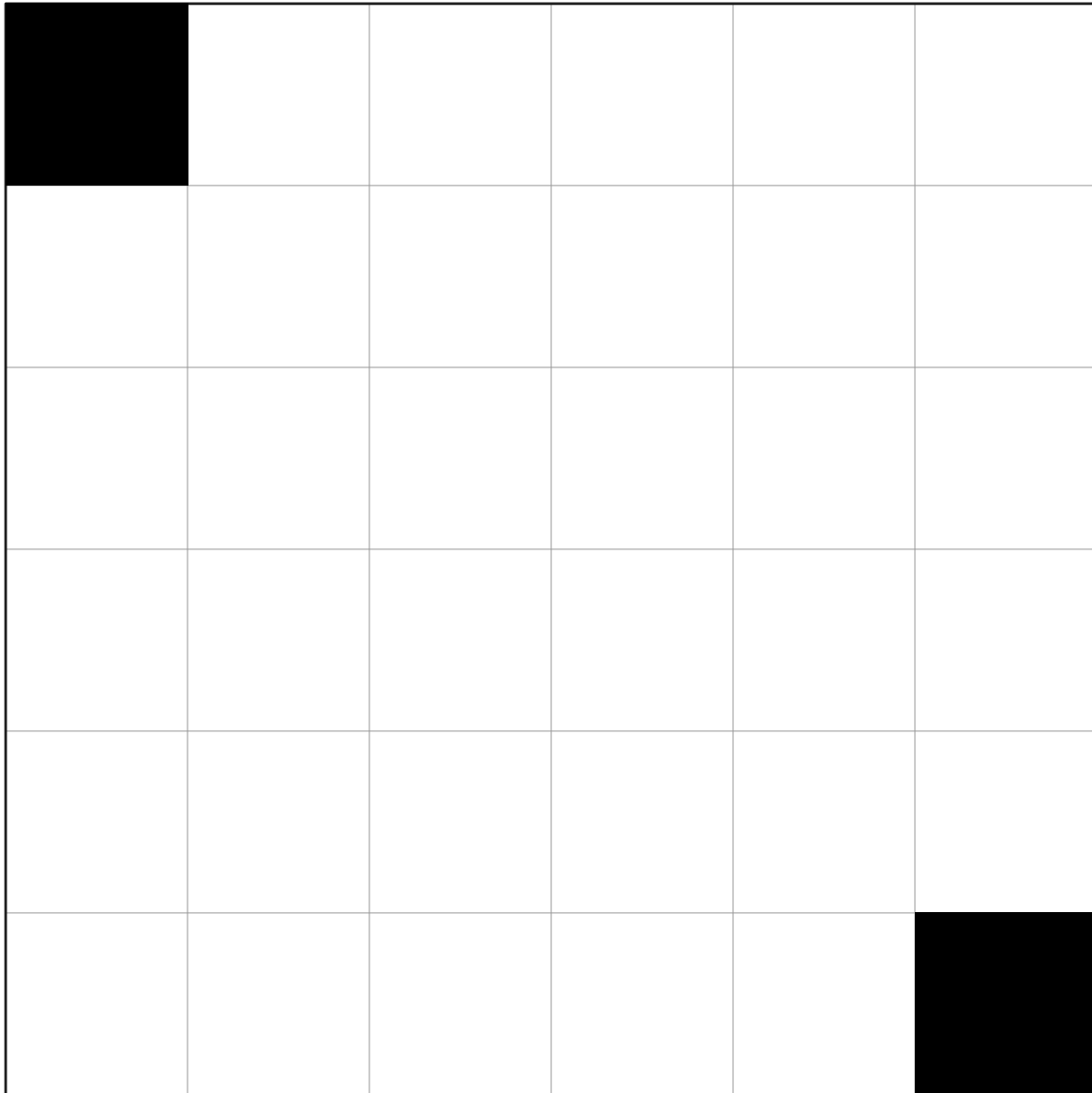
Problem 6: Choose two squares of different shades yourself and mark them black on a small board below. On the big board from Problem 4, put black squares in the same places and try to cover the rest with dominoes. Draw the dominoes on the small board. Do this for three different choices. Try hard to choose a pair that cannot be covered.

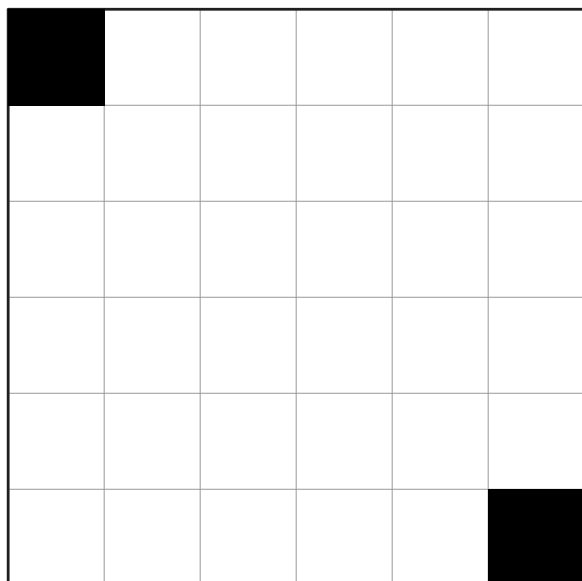


Problem 7: Draw a path on the big board: start in any square, move each step to a square that shares a side with the one you are in, never diagonally, visit every square exactly once, and finish in a square that shares a side with your starting square, so that the path closes into a loop. Then put two black squares on squares of different shades. Cover all the other squares with dominoes placed along your loop, each domino covering two squares that are next to each other on the loop. Record on the small boards: draw the loop, the black squares, and the dominoes. Do this for two different pairs of black squares.

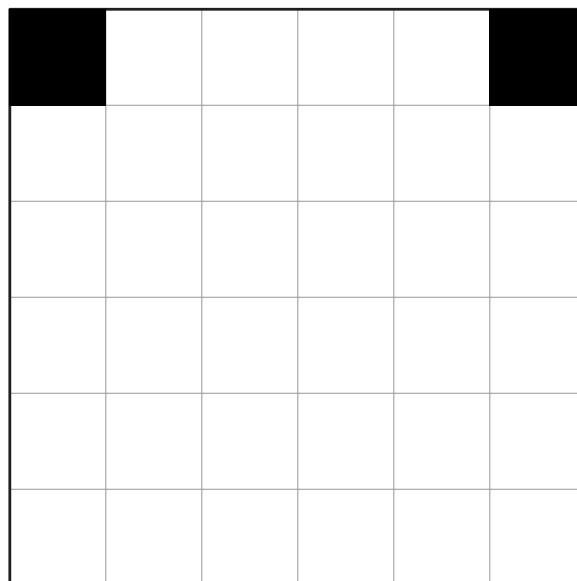


Problem 8: On this 6-by-6 board the two black squares are on opposite corners. Try to cover all the other squares with dominoes. If you can, draw the dominoes on the first small board on the next page. If you cannot, shade the first small board like a checkerboard and write under it how many shaded and how many unshaded squares are not black. Then move one black square to the other corner on the same side, try again, and record on the second small board in the same way.



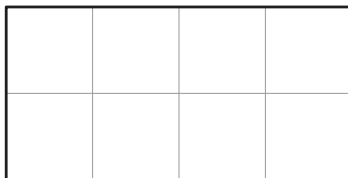
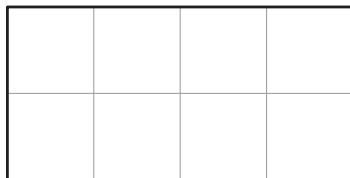
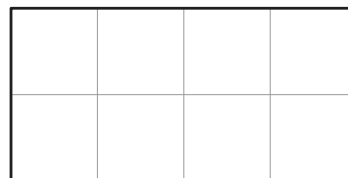
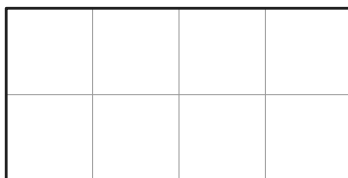
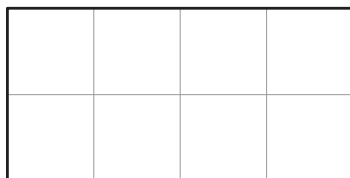
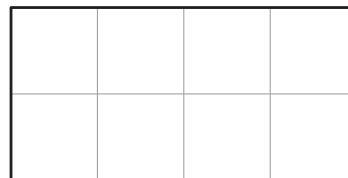
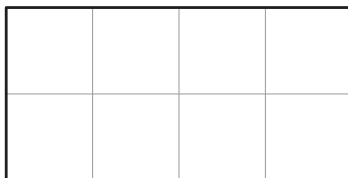
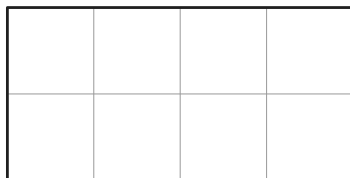


black squares on opposite corners
shaded _____ unshaded _____

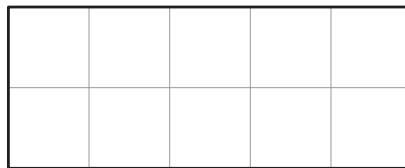
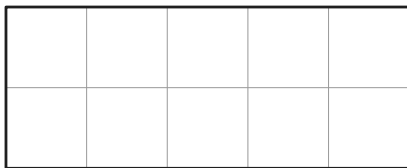
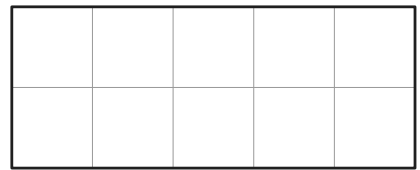
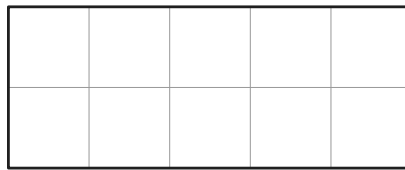
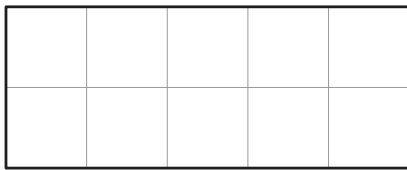
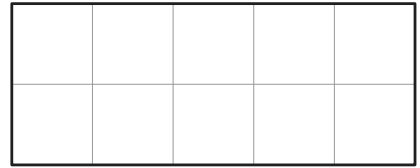
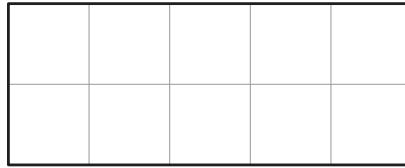
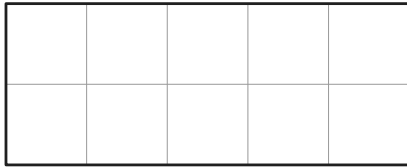
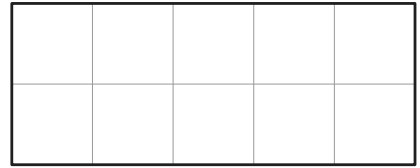
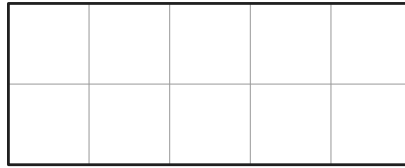
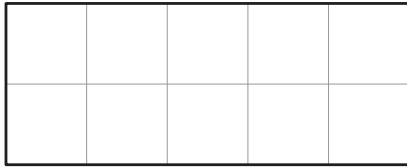


black squares on corners of the same side
shaded _____ unshaded _____

Problem 9: Find every different way to cover the 2-by-4 board with dominoes. Draw each way on its own small board. Then do the same for the 2-by-5 board on the next page. Write the number of ways for each board size in the blanks on the next page.



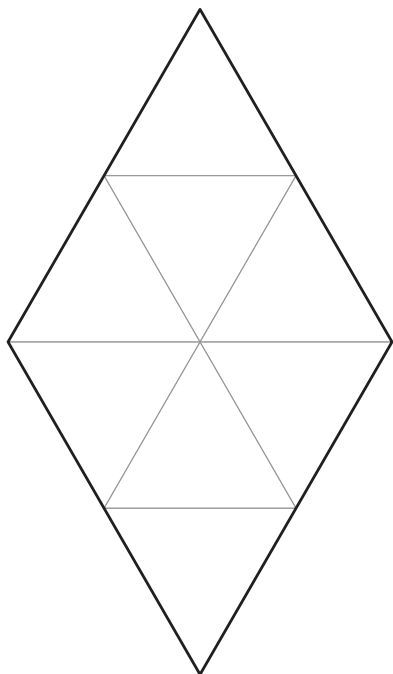
Problem 9: (continued) Every different way to cover the 2-by-5 board.



Number of ways: 2-by-2 _____ 2-by-3 _____ 2-by-4 _____ 2-by-5 _____

Write the number of ways you expect for the 2-by-6 board, then say to the adult how you got it: _____

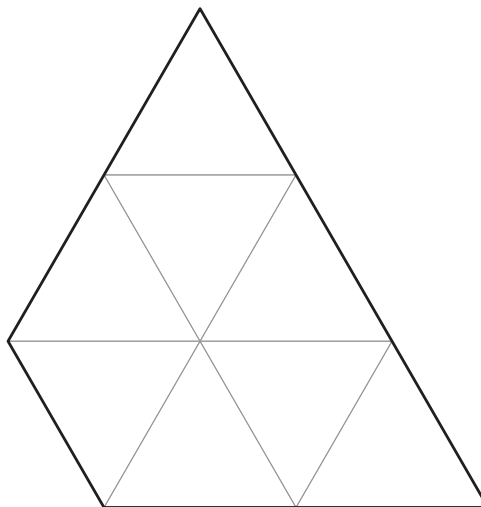
Problem 10: Try to cover each shape with blue rhombi only. A rhombus must lie exactly on two triangles of the picture, and rhombi may not overlap. If you cover the whole shape, draw a check in the box; if you cannot, draw an X. Then shade every triangle that points up, and write under the shape how many triangles are shaded and how many are not.



D



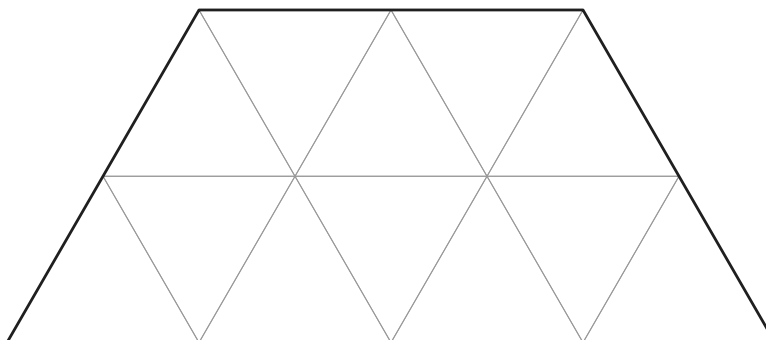
shaded _____ unshaded _____



E



shaded _____ unshaded _____



F



shaded _____ unshaded _____